Current-main formal target static audit

Crypto-autoresearcher commit: `4239648a889fe97d976a99862d178ba50324dc6e`. **Result:** `integrity_fail`. The 125 current formal target specs contain two duplicate task-ID groups. Every declared source and theorem file is readable from this exact Git tree. This is a source inventory, not a Lean build or scientific review.

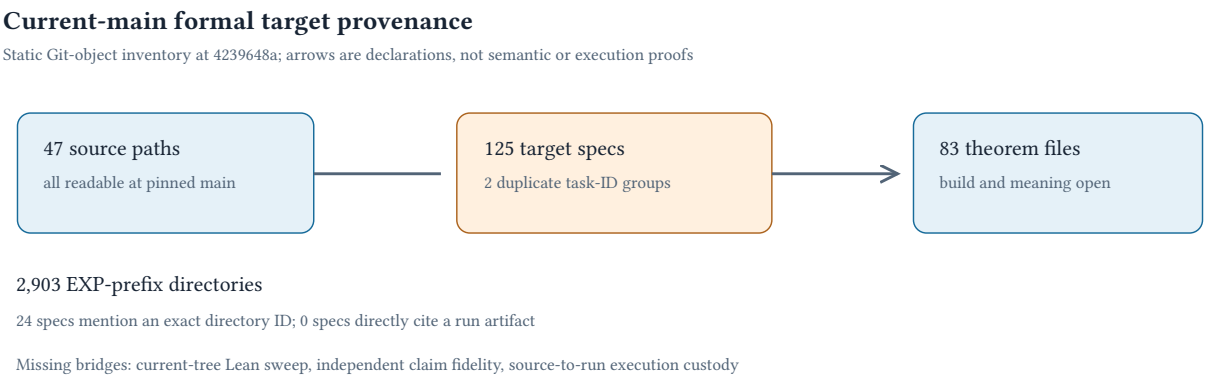

Figure 1: Declared provenance topology at the pinned commit. Dashed obligations in the prose are not established by these static links.

Counts and collision witnesses

The target set declares 83 distinct theorem files and 47 distinct source paths. It contains 125 unique claim IDs and 125 unique theorem names. The 2,903 top-level `EXP-*` directory names are not a run-validity count. Only 24 specs mention an exact existing experiment directory ID, zero specs cite a run explicitly, and zero declared source paths point into `/runs/`.

`TASK-20261104-df29f3` occurs in `binary-semaev-repair-20261105.yaml` and `closure-two-sided-certificates-20261104.yaml`. `TASK-20261105-9e5cbb` occurs in `boolean-lastfall-upper-20261105.yaml` and `boolean-normalization-repair-20261105.yaml`. The earlier 87-target builder requires unique task IDs and stops on this current tree; the new auditor's `--require-clean` mode exits 1. Its repeat output was byte-identical. As a positive control, the same strict mode passed on the earlier pinned 87-target tree with no collisions.

Provenance and limits

The primary repository sources are `formal/targets/README.md`, the pinned 2026-10-25 formal target provenance report, and `problems/formal-target-semantic-fidelity-unreviewed.md`. The replayable `audit.py` reads Git objects, validates schema and declared paths, computes SHA-256 values and emits `inventory.json` (SHA-256 `27cab7302451206d1204d4c4f92382519e1e156481782ea3bede59c399d73757`). Source-status labels are copied from specs; they are not reviewer verdicts.

No frozen target or receipt was edited. The next step is to identify each target's real task owner, then make a versioned correction that preserves the old receipt bindings. A current-tree Lean build and per-claim semantic and run-custody reviews remain open. This audit changes no performance graph or dashboard value.